

Time: 30 minutes Marks: 15

Question 1

How much energy is stored in a battery, if it can supply a 5V dc pump with 2m A constant current for one week?

- a) Energy in Wh, (w_h)
b) Energy in kJ, (w_j)

$$W = i V t$$
$$= 2 \mu \times 5 \times 7 \times 24 = 1.68 \text{ Wh} \quad \checkmark$$

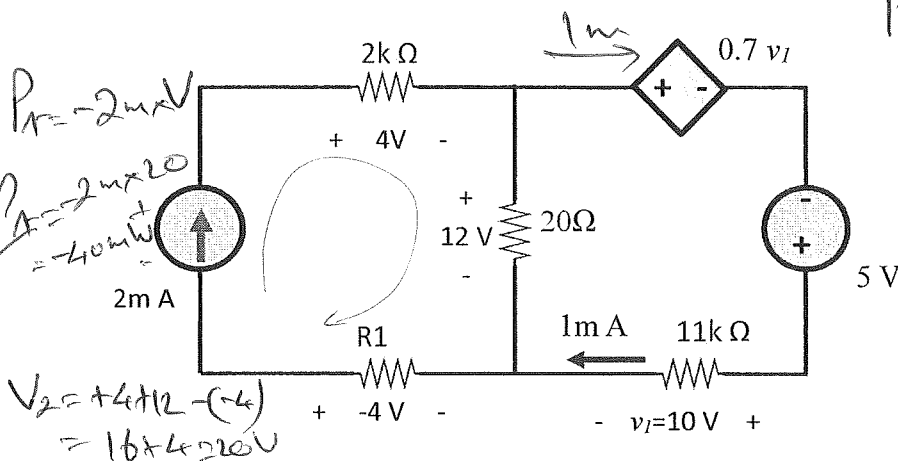
$$\omega = 1.68 \times 60 \times 60 = 6.05 \text{ kJ} \checkmark$$

Question 2

Calculate the power associated with

- 2m A current source, (P_A)
- Dependable source, (P_D)

-40 W (40 W delivered)



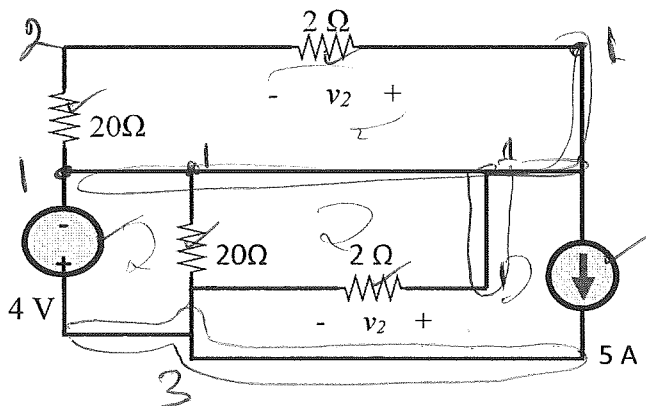
$$P_D = 1 \text{ m} \times 0.7 \text{ V}$$

$$= 1 \text{ m} \times 0.7 \times 10$$

$$= \underline{7 \text{ mW}} \checkmark$$

Question 3

Identify the number of nodes, (n), branches, (b) and independent loops (l), in the circuit shown below.

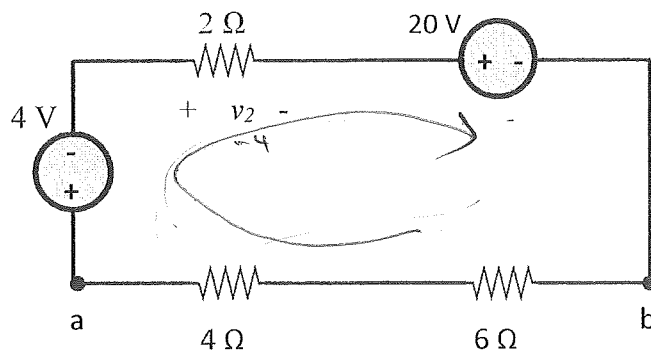


$$\begin{aligned} n &= 3 \quad \checkmark \\ b &= 6 \quad \checkmark \\ L &= b + 1 - n \\ &= 7 - 3 = 4 \quad \checkmark \end{aligned}$$

Question 4

Determine the voltages:

- a) v_2 and
- b) v_{ab}



$$v_2 = -(20+4) \times \frac{2}{2+4+6}$$

$$= -24 \times \frac{1}{6}$$

$$= -4V \quad \checkmark$$

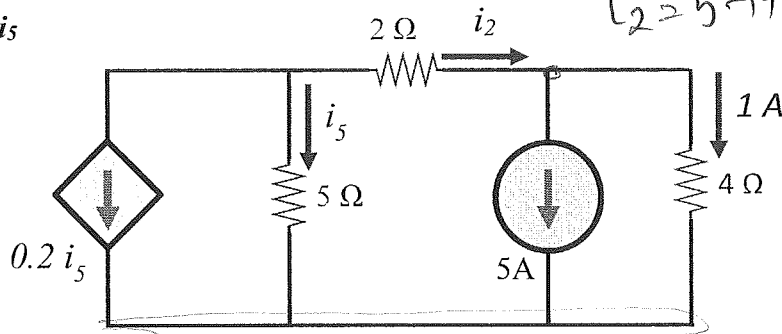
$$V_{ab} = 24 \left(\frac{4+6}{12} \right)$$

$$= 20V \quad \checkmark$$

Question 5

Calculate the currents:

- a) i_2 and
- b) i_5



$$i_2 = 5 + 1 = 6A \quad \checkmark$$

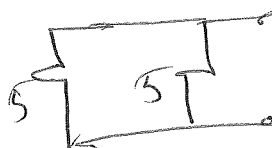
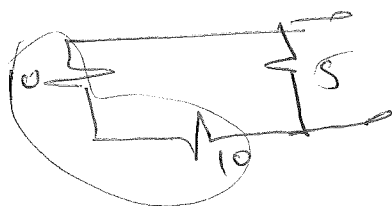
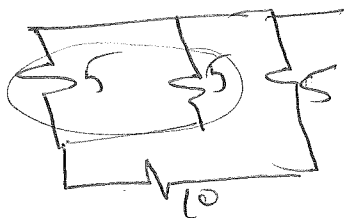
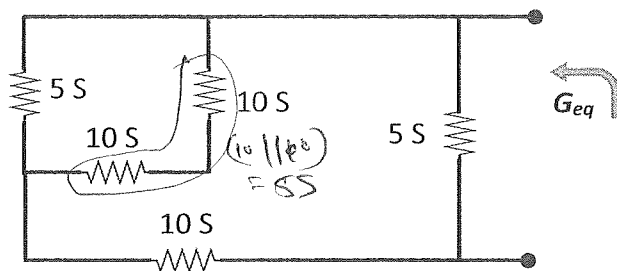
$$0.2i_5 + i_5 + 5 + 1 = 0$$

$$1.2i_5 = -6$$

$$i_5 = \frac{-6}{1.2} = -5A \quad \checkmark$$

Question 6

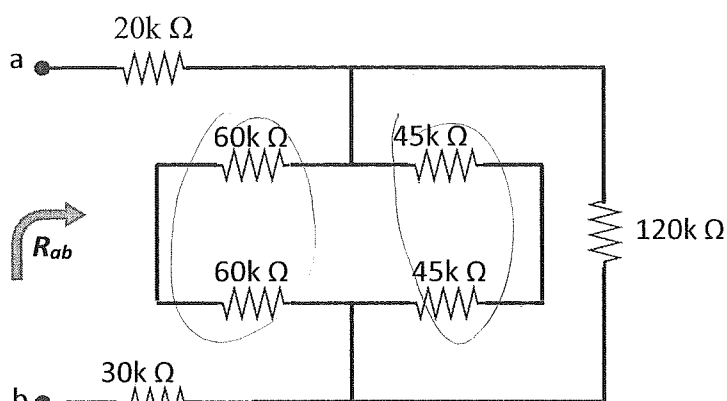
What is the equivalent conductance, G_{eq} , of the circuit shown below?



$$G_{eq} = (5 + 5) = 10 \text{ S}$$

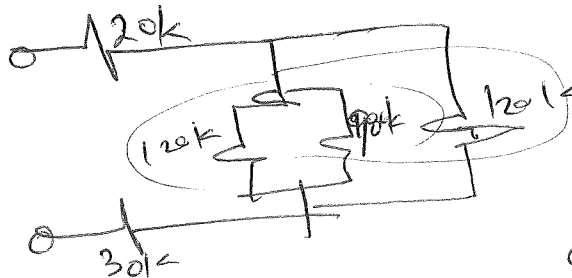
Question 7

What is the equivalent resistance, R_{ab} , across terminal "ab"?

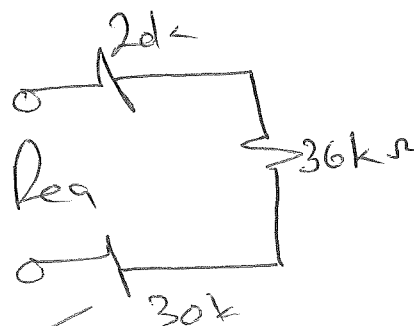


$$(60 \text{ k} + 60 \text{ k}) = 120 \text{ k}$$

$$(45 \text{ k} + 45 \text{ k}) = 90 \text{ k}$$



$$R_p = \frac{1}{120} + \frac{1}{120} + \frac{1}{90} = \frac{1}{60} + \frac{1}{90} = \frac{60 + 40}{60 \times 90} = \frac{100}{5400} = \frac{1}{54} \Rightarrow 54 \text{ k}$$



$$R_{eq} = 20 \text{ k} + 36 \text{ k} + 30 \text{ k}$$

$$= 86 \text{ k} \Omega$$

Question

Question	Answer	Question	Answer
1a)	$w_h =$	1b)	$w_J =$
2a)	$P_A =$	2b)	$P_D =$
3a)	$n =$	3b)	$b =$
3c)	$l =$		
4a)	$v_2 =$	4b)	$v_{ab} =$
5a)	$i_2 =$	5b)	$i_5 =$
6a)	$G_{eq} =$	6b)	$R_{eq} =$
7a)		7b)	